Heavens Light is Our Guide

Rajshahi University of Engineering and Technology
Department of Electrical & Computer Engineering

Course Title

Digital Techniques Sessional

Course No: ECE 2112

Lab Report -02

Name of the Experiment: Simplification of Boolean Expressions Using Boolean Algebra and Karnaugh Map, and Verification Using Logic Circuit Simulation

Submitted By:

Uddipon Chowdhury
Roll: 2410019
Registration No: 1072

Submitted To:

Mst. Mazeda Noor Tasnim
Lecturer
ECE, RUET

Problem 1

Problem Statement

Simplify the Boolean expression $F(A, B, C) = \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$ using Boolean algebra and the Karnaugh map, and verify both the original and the simplified circuits through logic simulation.

Objective

The goal is to reduce a three-variable sum-of-minterms expression to its minimal sum-of-products form, confirm that reduction with a Karnaugh map, and check that the original and simplified circuits behave identically by building both in Logisim.

Algebraic Simplification

$$\begin{aligned}
F(A, B, C) &= \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC \\
&= \overline{A}BC + ABC + A\overline{B}C + ABC + AB\overline{C} + ABC \\
&= BC(\overline{A} + A) + AC(\overline{B} + B) + AB(\overline{C} + C) \\
&= BC + AC + AB \\
&= AB + AC + BC
\end{aligned}$$

Truth Table

A	B	C	$F = AB + AC + BC$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Table 1: Truth table for $F = AB + AC + BC$

Karnaugh Map

$A \backslash BC$	**00**	**01**	**11**	**10**
0	0	0	1	0
1	0	1	1	1

Table 2: K-map for $F(A, B, C) = \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$

Three pairs of adjacent 1-cells can be grouped: the pair at $BC = 11$ (covering both rows) gives BC; the pair with $A = 1,\ C = 1$ (i.e. $BC = 01, 11$) gives AC; and the pair with $A = 1,\ B = 1$ (i.e. $BC = 11, 10$) gives AB. Combining these three groups produces the minimized expression $F = AB + AC + BC$, which agrees with the algebraic result obtained above.

Circuit Diagram

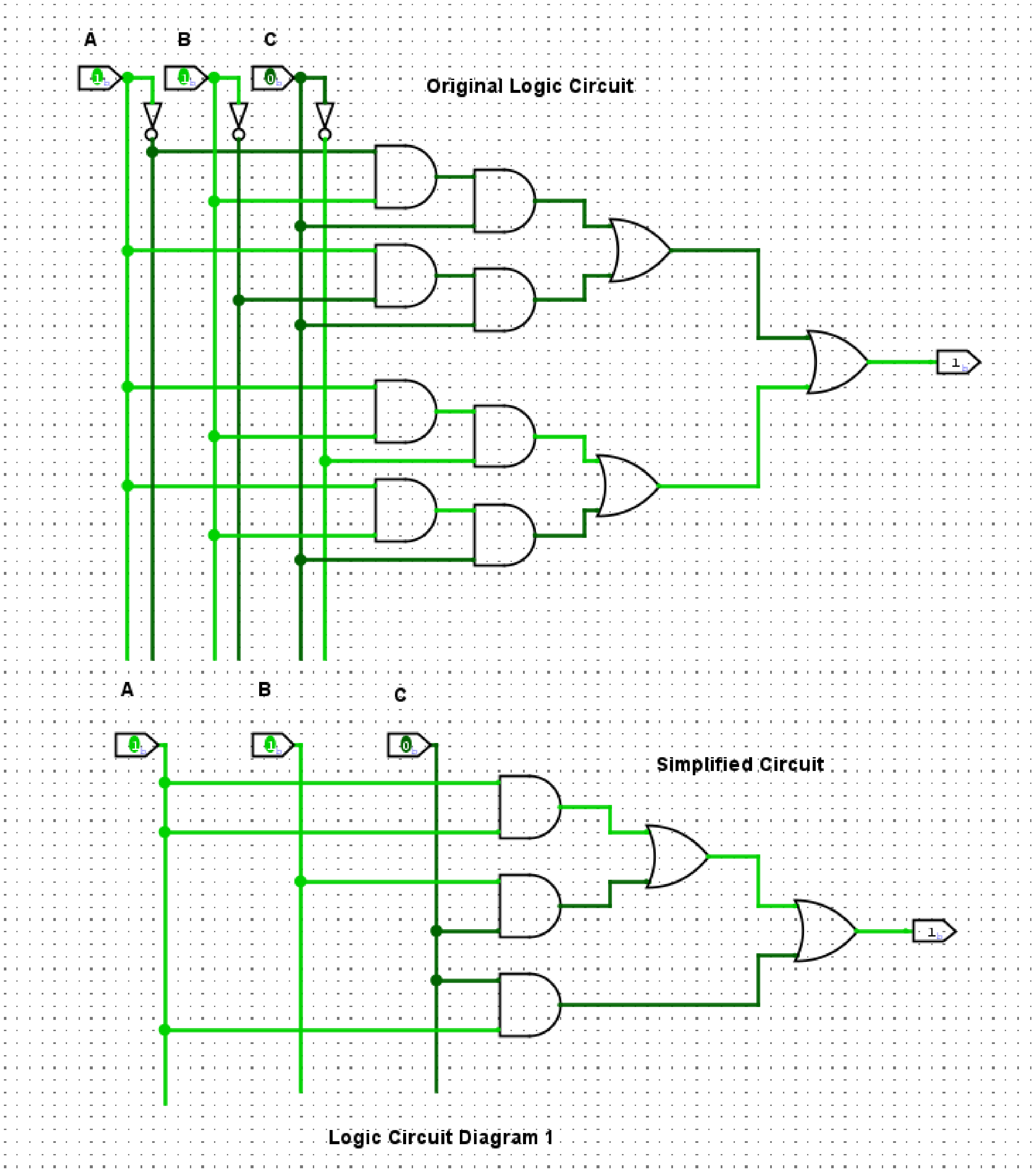

Figure 1: Original circuit (four 3-input AND gates plus a 3-input OR gate, realizing $\overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$) and simplified circuit (three 2-input AND gates plus an OR gate, realizing $AB + AC + BC$), simulated in Logisim

Discussion

The original circuit implements the expression exactly as written: it needs four 3-input AND gates, one for each product term, along with inverters to generate $\overline{A}$, $\overline{B}$, and $\overline{C}$

wherever they appear, and a 3-input OR gate to combine the terms. The simplified circuit, in contrast, realizes $AB + AC + BC$ using only three 2-input AND gates feeding a single OR gate, and it needs no inverters at all. Simulating both versions in Logisim for every possible input combination should produce identical outputs, confirming that the algebraic and Karnaugh-map simplifications are correct. Overall, the simplified circuit is clearly the more efficient design: it uses fewer gates, fewer total gate inputs, and eliminates the inverters entirely.

Conclusion

The expression $F = \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$ was successfully reduced to $F = AB + AC + BC$ through Boolean algebra and confirmed independently with a Karnaugh map. Logisim simulation of both the original and the simplified circuits is expected to give identical outputs across all input combinations, verifying that the simplification is correct.

Problem 2

Problem Statement

Simplify the Boolean expression $F(A, B, C) = A(A + B)(A + B + C)$ using Boolean algebra and the Karnaugh map, and verify both the original and the simplified circuits through logic simulation.

Objective

The aim is to reduce a three-variable product-of-sums-type expression to its simplest possible form, confirm that reduction using a Karnaugh map, and check the equivalence of the original and simplified circuits in Logisim.

Algebraic Simplification

$$\begin{aligned}
F(A, B, C) &= A(A + B)(A + B + C) \\
&= (A + AB)(A + B + C) \\
&= A(A + B + C) + AB(A + B + C) \\
&= A + AB + AC + AB + AB + ABC \\
&= A + AB + AC + ABC \\
&= A(1 + B + C + BC) \\
&= A
\end{aligned}$$

Truth Table

A	B	C	$F = A$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Table 3: Truth table for $F = A$

Karnaugh Map

A \ BC	**00**	**01**	**11**	**10**
0	0	0	0	0
1	1	1	1	1

Table 4: K-map for $F(A, B, C) = A(A + B)(A + B + C)$

All four cells in the $A = 1$ row are 1, while the entire $A = 0$ row is 0. These four adjacent cells form a single group spanning the whole row, eliminating both B and C and leaving only $F = A$ – exactly matching the algebraic simplification above.

Circuit Diagram

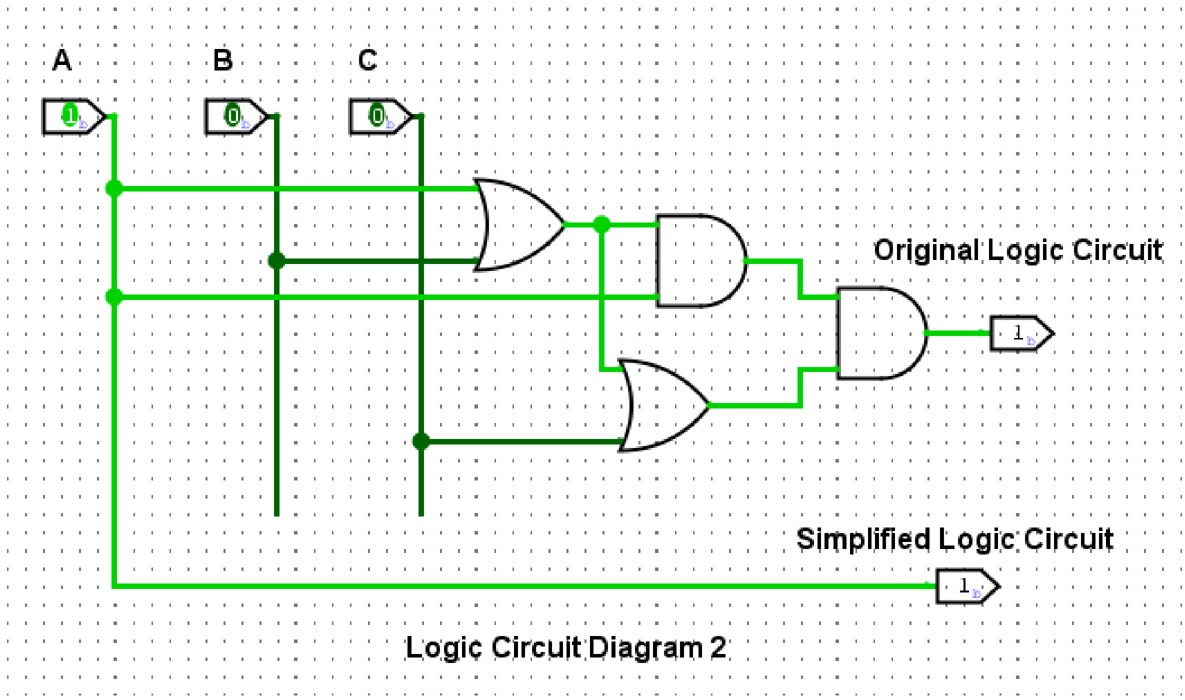

Figure 2: Original circuit (two OR gates realizing $A + B$ and $A + B + C$, combined with A through an AND gate) and simplified circuit (a direct connection representing $F = A$), simulated in Logisim

Discussion

The original circuit first forms $A + B$ with one OR gate, extends that result to $A + B + C$ with a second OR gate, and then ANDs both results with A itself. Since A is common to every term in the expression, all of the higher-order terms are absorbed, and the whole network collapses to the single literal A; the simplified circuit is therefore nothing more than a wire connecting input A directly to the output, with no gates required. This is a textbook example of the absorption law $A + AB = A$ being applied twice in succession. Testing both circuits in Logisim for every combination of A, B, and C should confirm that the output always equals A, irrespective of B and C.

Conclusion

The expression $F = A(A + B)(A + B + C)$ was shown to reduce to $F = A$ through repeated application of the absorption law. This result is confirmed by the Karnaugh map and is expected to be validated by Logisim simulation, which should show that B and C have no effect on the output.

Problem 3

Problem Statement

Simplify the Boolean expression $F(A, B, C) = (A + (BC)')' (AB + ABC)$ using Boolean algebra and the Karnaugh map, and verify both the original and the simplified circuits through logic simulation.

Objective

The objective is to reduce a mixed AND-OR-complement expression using De Morgan's theorem together with Boolean algebra, confirm the result with a Karnaugh map, and use Logisim to verify that the simplified circuit behaves as a constant.

Algebraic Simplification

$$\begin{aligned}
F(A, B, C) &= \overline{A + \overline{BC}}\,(AB + ABC) \\
&= \overline{A} \cdot BC \cdot (AB + ABC) \qquad \text{(De Morgan's theorem)} \\
&= \overline{A}BC \cdot AB(1 + C) \\
&= \overline{A}BC \cdot AB \\
&= (A\overline{A})\,BC\,B \\
&= 0
\end{aligned}$$

Truth Table

A	B	C	$F = 0$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Table 5: Truth table for $F = 0$

Karnaugh Map

A \ BC	**00**	**01**	**11**	**10**
0	0	0	0	0
1	0	0	0	0

Table 6: K-map for $F(A,B,C) = \overline{A+\overline{BC}}\,(AB+ABC)$

Every cell of the map is 0, so there are no 1-cells available to group. A Karnaugh map with no populated cells represents the constant function $F = 0$, which agrees with the algebraic result.

Circuit Diagram

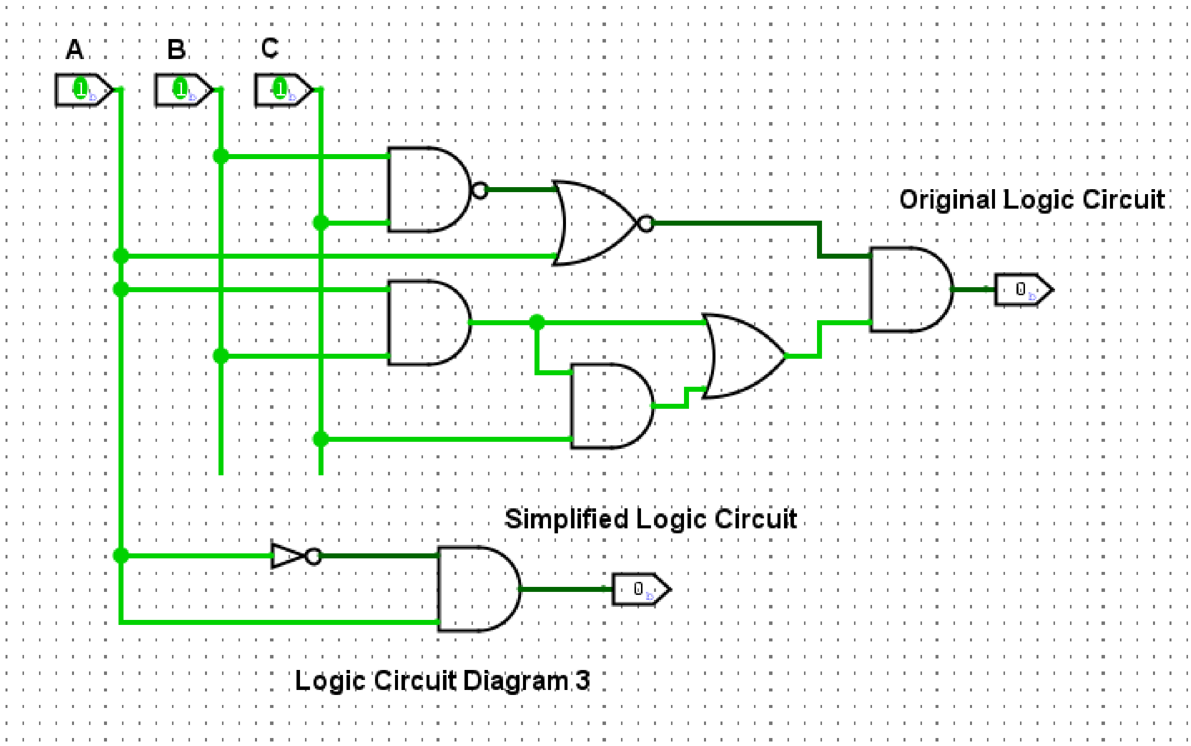

Figure 3: Original circuit realizing $(A + (BC)')'\,(AB+ABC)$ using NAND-type and OR/AND stages, and simplified circuit showing the output permanently at logic 0, simulated in Logisim

Discussion

Applying De Morgan's theorem to $\overline{A+\overline{BC}}$ converts it into $\overline{A}\cdot BC$; the second factor, $AB + ABC$, then simplifies to AB by absorption, since $AB + ABC = AB(1 + C) = AB$.

Multiplying $\overline{A}BC$ by AB introduces the term $A \cdot \overline{A}$, which is always 0, so the entire expression is forced to 0 regardless of the values of A, B, and C. Cycling through all eight input combinations on the original circuit in Logisim should show the output remaining at logic 0 throughout, exactly as predicted by the all-zero Karnaugh map.

Conclusion

The expression $F = (A + (BC)')' \, (AB + ABC)$ was shown to be identically zero for every input combination. This was verified algebraically and confirmed by an empty Karnaugh map; Logisim simulation is expected to validate this by showing a constant logic-0 output regardless of A, B, and C.

Problem 4

Problem Statement

Simplify the Boolean expression $F(A, B) = \left[\overline{B}(A + B) + (A + B)(A + \overline{B})\right] \overline{B}$ using Boolean algebra and the Karnaugh map, and verify both the original and the simplified circuits through logic simulation.

Objective

The goal is to reduce a two-variable expression built from nested OR and AND terms to its minimal form, confirm the reduction using a Karnaugh map, and verify the equivalence of the original and simplified circuits in Logisim.

Algebraic Simplification

$$\begin{aligned} F(A, B) &= \left[\overline{B}(A + B) + (A + B)(A + \overline{B})\right] \overline{B} \\ &= (A\overline{B} + B\overline{B} + AA + A\overline{B} + A\overline{B} + B\overline{B})\overline{B} \\ &= (A + A\overline{B} + A\overline{B} + A\overline{B})\overline{B} \\ &= A(1 + \overline{B} + \overline{B} + \overline{B})\,\overline{B} \\ &= A\overline{B} \end{aligned}$$

Truth Table

A	B	$F = A\overline{B}$
0	0	0
0	1	0
1	0	1
1	1	0

Table 7: Truth table for $F = A\overline{B}$

Karnaugh Map

$A \backslash B$	**0**	**1**
0	0	0
1	1	0

Table 8: K-map for $F(A, B) = \left[\overline{B}(A + B) + (A + B)(A + \overline{B})\right] \overline{B}$

Only a single cell, at $A = 1,\ B = 0$, is populated. A lone 1-cell cannot be paired with any adjacent cell, so it is read directly as the product of its coordinate literals, giving $F = A\overline{B}$ – matching the algebraic result.

Circuit Diagram

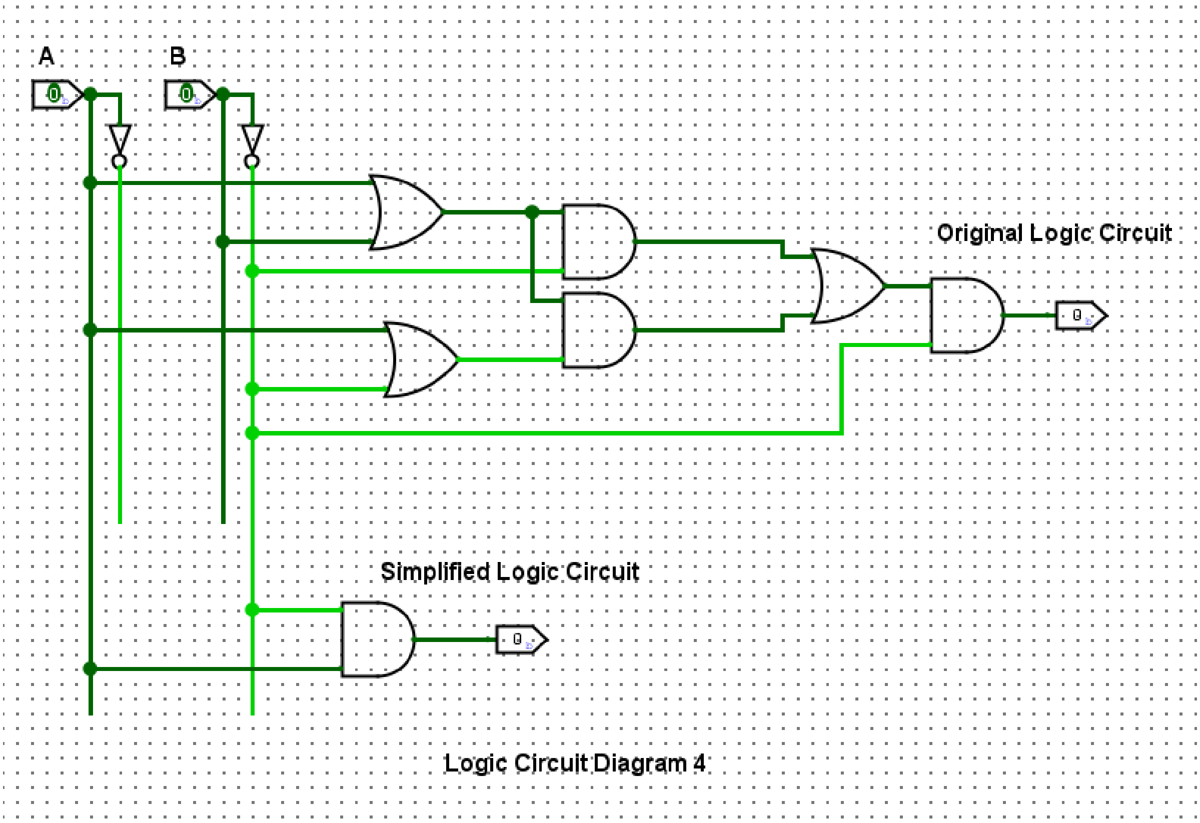

Figure 4: Original circuit realizing $\left[\overline{B}(A + B) + (A + B)(A + \overline{B})\right] \overline{B}$ using OR, AND, and NOT stages, and simplified circuit reduced to a single AND gate realizing $A\overline{B}$, simulated in Logisim

Discussion

The original expression can be simplified by F = $A\overline{B}$.

Thus, the original multi-gate network is functionally equivalent to a simplified implementation of $A\overline{B}$, requiring an AND operation with the B input complemented. Testing both circuits in Logisim for all four combinations of A and B should confirm that the output is 1 only when $A = 1$ and $B = 0$, matching the truth table and Karnaugh map.

Conclusion

The expression $F = \left[\overline{B}(A+B) + (A+B)(A+\overline{B})\right]\overline{B}$ was successfully reduced to $F = A\overline{B}$ using Boolean algebra and confirmed with a Karnaugh map. Logisim simulation of the original and simplified circuits is expected to show identical outputs across all input combinations.

References

[1] M. M. Mano, *Digital Design*, 5th ed. Upper Saddle River, NJ: Pearson, 2013.

[2] R. J. Tocci, N. S. Widmer, and G. L. Moss, *Digital Systems: Principles and Applications*, 11th ed. Upper Saddle River, NJ: Pearson, 2011.